

US Patent & Trademark Office

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United States Patent Application Publication

20250274122

Kind Code

A1

Publication Date

August 28, 2025

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INPUT BUFFER CIRCUIT HAVING VARIABLE THRESHOLD CIRCUITS

Abstract

An integrated circuit includes an asymmetry upper threshold circuit outputting an upper-branch enabling signal, an asymmetry lower threshold circuit outputting a lower-branch enabling signal, and a control circuit which changes an output voltage when the logic levels of the upper-branch enabling signal and the lower-branch enabling signal are changed consecutively. The upper-branch enabling signal is set to a first upper-branch logic level when an input voltage received by the integrated circuit rises across a main upper threshold, and to a second upper-branch logic level when the input voltage falls across a supplementary upper threshold which is higher than the main upper threshold. The lower-branch enabling signal is set to a first lower-branch logic level when the input voltage falls across a main lower threshold, and to a second lower-branch logic level when the input voltage rises across a supplementary lower threshold which is lower than the main lower threshold.

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Family ID: 1000007805755

Appl. No.: 18/587077

Filed: February 26, 2024

Publication Classification

Int. Cl.: H03K19/00 (20060101); H03K17/687 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is related to U.S. application Ser. No. 17/338,107, filed Jun. 3, 2021, and titled, "Input Buffer Circuit," which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] The recent trend in miniaturizing integrated circuits (ICs) has resulted in smaller devices which consume less power yet provide more functionality at higher speeds. The miniaturization process has also resulted in stricter design and manufacturing specifications as well as reliability challenges. Various electronic design automation (EDA) tools generate, optimize and verify standard cell layout designs for integrated circuits while ensuring that the standard cell layout design and manufacturing specifications are met.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1A is a schematic diagram of an input buffer circuit having two threshold circuits and one control circuit, in accordance with some embodiments.

[0005] FIG. 1B is a schematic diagram of an implementation of the input buffer circuit of FIG. 1A in combination with level shifters, in accordance with some embodiments.

[0006] FIG. 1C is a schematic diagram of an input buffer circuit having the control circuit implemented with switches, in accordance with some embodiments.

[0007] FIGS. 2A-2F are diagrams of waveforms of signals in connection with the input buffer circuit of FIG. 1B, in accordance with some embodiments.

[0008] FIGS. 3A-3F are diagrams of waveforms of signals in connection with the input buffer circuit in FIG. 1C, in accordance with some embodiments.

[0009] FIGS. 4A-4B are schematic diagrams of input buffer circuits having variable thresholds in threshold circuits, in accordance with some embodiments.

[0010] FIGS. 5A-5F are diagrams of waveforms of signals at various nodes of an improved input buffer circuit, in accordance with some embodiments.

[0011] FIG. 6 is a schematic diagram of an input buffer circuit which has threshold circuits controlled with threshold-selection signals, in accordance with some embodiments.

[0012] FIG. 7 is a flowchart of a method of generating an output voltage signal from an input voltage signal, in accordance with some embodiments.

[0013] FIG. 8A is a schematic diagram of an asymmetry upper threshold circuit, in accordance with some embodiments.

[0014] FIG. 8B is a schematic diagram of an asymmetry lower threshold circuit, in accordance with some embodiments.

DETAILED DESCRIPTION

[0015] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components,

values, operations, materials, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0016] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for case of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0017] An input buffer circuit couples an input voltage signal from a transmitting circuit in a first power domain to a receiving circuit in a second power domain. The input buffer circuit receives the input voltage signal and generates an output voltage signal. The input buffer circuit has a lower threshold voltage and an upper threshold voltage. In some embodiments, an upper-branch enabling signal is generated based a comparison between the input voltage signal and the upper threshold voltage, and a lower-branch enabling signal is generated based a comparison between the input voltage signal and the lower threshold voltage. In some embodiments, based on the upper-branch enabling signal and the lower-branch enabling signal, the output voltage signal changes between a first voltage level and a second voltage level. When the input voltage signal rises and consecutively crosses the lower threshold voltage and then the upper threshold voltage, the output voltage signal changes from the first voltage level to the second voltage level. When the input voltage signal falls and consecutively crosses the upper threshold voltage and then the lower threshold voltage, the output voltage signal changes from the second voltage level to the first voltage level.

[0018] FIG. 1A is a schematic diagram of an input buffer circuit **100** having two threshold circuits and one control circuit, in accordance with some embodiments. The input buffer circuit **100** includes an asymmetry upper threshold circuit **110**, an asymmetry lower threshold circuit **120**, and a control circuit **130**. The asymmetry upper threshold circuit **110** and the asymmetry lower threshold circuit **120** are configured to receive an input voltage signal PAD at an input node **102** of the input buffer circuit **100**. The asymmetry upper threshold circuit **110** is configured to generate an upper-branch enabling signal EN.sub.up based on a comparison between the input voltage signal PAD and the upper threshold voltage VTH. In some embodiments, if the input voltage signal PAD is larger than the upper threshold voltage VTH, then the upper-branch enabling signal EN.sub.up generated by the asymmetry upper threshold circuit **110** is set to be logic TRUE. The asymmetry lower threshold circuit **120** is configured to generate a lower-branch enabling signal EN.sub.dn based on a comparison between the input voltage signal PAD and the lower threshold voltage VTL. In some embodiments, if the input voltage signal PAD is smaller than the lower threshold voltage VTL, then the lower-branch enabling signal EN.sub.dn generated by the asymmetry lower threshold circuit **120** is set to be logic TRUE. The upper-branch enabling signal EN.sub.up and the lower-branch enabling signal EN.sub.dn are coupled to the control circuit **130**. The control circuit **130** is configured to generate an output voltage signal Vout at the output node **108** of the control circuit **130** based on the upper-branch enabling signal EN.sub.up and the lower-branch enabling signal EN.sub.dn.

[0019] In operation, the input voltage signal PAD at the input node **102** is provided by electronic

circuits in a first power domain, and the output voltage signal Vout generated by the input buffer circuit **100** at the output node **108** is coupled to electronic circuits in a second power domain. In some embodiments, the electronic circuits in the first power domain are connected between the power supply voltages VDDH and VSS, and the electronic circuits in the second power domain are connected between the power supply voltages VDDL and VSS. In some embodiments, the power supply voltage VSS is connected to the common ground, and the power supply voltage VDDH in the first power domain is 2.5V (or 3.3V) while the power supply voltage VDDL in the second power domain is 1.8V. In some embodiments, the power supply voltage VSS is connected to common ground, and the power supply voltage VDDH in the first power domain is 1.8V (or 2.5V) while the power supply voltage VDDL in the second power domain is 1.2V. In some embodiments, the power supply voltage VDDH is higher than 3.3V. In some embodiments, the power supply voltage VDDL is smaller than 1.2 V. Other examples of the power supply voltage VDDH and the power supply voltage VDDL are within the contemplated scope of the present disclosure. During operation, the voltage levels of the input voltage signal PAD at the input node **102** of the input buffer circuit **100** generally are in a range from VSS to VDDH, and the output voltage signal Vout generated at the output node **108** of the input buffer circuit **100** is in a range from VSS to VDDL.

[0020] The output voltage signal Vout at the output node **108** of the input buffer circuit **100** changes in response to the input voltage signal PAD at the input node **102**. The input voltage signal PAD follows a trapezoid waveform: the input voltage signal PAD rises from voltage VSS to voltage VDDH and stays at the voltage VDDH for some time; then the input voltage signal PAD falls from voltage VDDH to voltage VSS. As the input voltage signal PAD is rising from voltage VSS and changing towards voltage VDDH, the output voltage signal Vout changes from voltage VSS to voltage VDDL at time $t+$, when the input voltage signal PAD crosses an upper threshold voltage VTH. The output voltage signal Vout stays at voltage VDDL, while the input voltage signal PAD reaches voltage VDDH and stays at voltage VDDH. The output voltage signal Vout changes from voltage VDDL to voltage VSS at time $t-$, when the input voltage signal PAD crosses a lower threshold voltage VTL, as the input voltage signal PAD is falling from voltage VDDH and changing towards voltage VSS. Because the signal waveform of the output voltage signal Vout changes within the range from VSS to VDDL, the output voltage signal Vout is a more suitable signal for the electronic circuits in the second power domain powered by the power supply voltages VDDL and VSS. In contrast, if the input voltage signal PAD is directly coupled to the electronic circuits in the second power domain, the peak voltage (such as VDDH) of the input voltage signal PAD may exceed the maximum durable voltage of the electronic circuits in the second power domain.

[0021] FIG. **1B** is a schematic diagram of an input buffer circuit **100B** having the control circuit implemented with switches, in accordance with some embodiments. In FIG. **1B**, the asymmetry upper threshold circuit **110** includes a high-side tracker **112** and an asymmetry upper threshold detector **114**. The high-side tracker **112** is configured to generate a tracking-up signal PADUP based on the input voltage signal PAD. The asymmetry upper threshold detector **114** is configured to receive the tracking-up signal PADUP from the high-side tracker **112** and to set a logic level of the upper-branch enabling signal ENUPB based on the tracking-up signal PADUP. In FIG. **1B**, the asymmetry lower threshold circuit **120** includes a low-side tracker **122** and an asymmetry lower threshold detector **124**. The low-side tracker **122** is configured to generate a tracking-down signal PADDN based on the input voltage signal PAD. The asymmetry lower threshold detector **124** is configured to receive the tracking-down signal PADDN from the low-side tracker **122** and to set a logic level of the lower-branch enabling signal ENDN based on the tracking-down signal PADDN.

[0022] In FIG. **1B**, the control circuit **130** includes a first switch **131**, a second switch **132**, and a regenerative circuit **135**. The first switch **131** is electrically connected between the upper supply voltage VDDH and a buffer output node BufOut. The second switch is **132** electrically connected between the buffer output node BufOut and the lower supply voltage VSS. The buffer output node

BufOut is at the electric connection between the first switch **131** and the second switch **132**. The regenerative circuit **135** is electrically coupled to the buffer output node BufOut. The regenerative circuit **135** maintains the voltage at the buffer output node BufOut when both the first switch **131** and the second switch **132** are at the disconnected state. The first switch **131** is controlled by the upper-branch enabling signal ENUPB received from the asymmetry upper threshold detector **114**. The second switch **132** is controlled by the lower-branch enabling signal ENDN received from the asymmetry lower threshold detector **124**. The buffer output node BufOut is coupled to the input terminal of the level shifter **140**. The level shifter **140** is connected to the power supply VDDH in the first power domain and the power supply VDDL in the second power domain. The voltage signal at the input terminal of the level shifter **140** is within the voltage range of the first power domain, but the voltage signal at the output terminal of the level shifter **140** is within the voltage range of the second power domain. The output terminal of the level shifter **140** is the output node **108** of the input buffer circuit **100B**. The output voltage signal Vout at the output node **108** of the input buffer circuit **100B** is within the voltage range of the second power domain. The operation of the input buffer circuit **100B** in FIG. **1B** is described with reference to FIGS. **2A-2F**.

[0023] FIGS. **2A-2F** are diagrams of waveforms of signals at the input node, the output node, and various other nodes in an input buffer circuit **100B** in FIG. **1B**, in accordance with some embodiments. FIG. **2A** is the waveform of the input voltage signal PAD at the input node **102**. FIG. **2B** is the waveform of the tracking-up signal PADUP at the output of the high-side tracker **112**. FIG. **2C** is the waveform of the tracking-down signal PADDN at the output of the low-side tracker **122**. FIG. **2D** is the waveform of the upper-branch enabling signal ENUPB at the output of the asymmetry upper threshold detector **114**. FIG. **2E** is the waveform of the lower-branch enabling signal ENDN at the output of the asymmetry lower threshold detector **124**. FIG. **2F** is the waveform of the voltage at the buffer output node BufOut.

[0024] In the example waveform of FIG. **2A**, during the time period from t_a to t_b , the input voltage signal PAD rises from voltage VSS to voltage VDDH. During the time period from t_b to t_c , the input voltage signal PAD remains at the voltage VDDH. During the time period from t_b to t_c , the input voltage signal PAD is maintained at voltage VDDH. During the time period from t_c to t_d , the input voltage signal PAD falls from voltage VDDH to voltage VSS. During the time period from t_a to t_b when the input voltage signal PAD is rising, the input voltage signal PAD crosses the lower threshold voltage VTL at time t_1 and crosses the upper threshold voltage VTH at time t_2 . During the time period from t_c to t_d when the input voltage signal PAD is falling, the input voltage signal PAD crosses the upper threshold voltage VTH at time t_3 and crosses the lower threshold voltage VTL at time t_4 .

[0025] In the example waveform of FIG. **2B**, the tracking-up signal PADUP at the output of the high-side tracker **112** follows the signal received at the input of the high-side tracker **112** if the signal received at the input is larger than a predetermined lower limiting voltage (such as VSSH), and the tracking-up signal PADUP is maintained at the predetermined lower limiting voltage (such as VSSH) if the signal received at the input is smaller than or equal to the predetermined lower limiting voltage.

[0026] In the example waveform of FIG. **2C**, the tracking-down signal PADDN at the output of the low-side tracker **122** follows the signal received at the input of the low-side tracker **122** if the signal received at the input is smaller than a predetermined upper limiting voltage (such as VDDL), and the tracking-down signal PADDN is maintained at the predetermined upper limiting voltage (such as VDDL) if the signal received at the input is larger than or equal to the predetermined upper limiting voltage.

[0027] In the example waveform of FIG. **2D**, the upper-branch enabling signal ENUPB at the output of the asymmetry upper threshold detector **114** is determined by comparing the tracking-up signal PADUP (received from the high-side tracker **112**) with the upper threshold voltage VTH. Before time t_2 , the tracking-up signal PADUP in FIG. **2B** is below the upper threshold voltage

VTH, and the upper-branch enabling signal ENUPB in FIG. 2D is at logic FALSE. From time t2 to time t3, the tracking-up signal PADUP in FIG. 2B is above the upper threshold voltage VTH, and the upper-branch enabling signal ENUPB in FIG. 2D is at logic TRUE. After time t3, the tracking-up signal PADUP in FIG. 2B is again below the upper threshold voltage VTH, and the upper-branch enabling signal ENUPB in FIG. 2D is at logic FALSE. In the input buffer circuit 100B of FIG. 1B, the logic TRUE of the upper-branch enabling signal ENUPB is implemented with a lower voltage level, while the logic FALSE of the upper-branch enabling signal ENUPB is implemented with a higher voltage level.

[0028] In the example waveform of FIG. 2E, the lower-branch enabling signal ENDN at the output of the asymmetry lower threshold detector 124 is determined by comparing the tracking-down signal PADDN (received from the low-side tracker 122) with the lower threshold voltage VTL. Before time t1, the tracking-down signal PADDN in FIG. 2C is below the lower threshold voltage VTL, and the lower-branch enabling signal ENDN in FIG. 2E is at logic TRUE. From time t1 to time t4, the tracking-down signal PADDN in FIG. 2C is above the lower threshold voltage VTL, and the lower-branch enabling signal ENDN in FIG. 2E is at logic FALSE. After time t4, the tracking-down signal PADDN in FIG. 2C is again below the lower threshold voltage VTL, and the lower-branch enabling signal ENDN in FIG. 2E is at logic TRUE. In the input buffer circuit 100B of FIG. 1B, the logic TRUE of the lower-branch enabling signal ENDN is implemented with a higher voltage level, while the logic FALSE of the lower-branch enabling signal ENDN is implemented with a lower voltage level.

[0029] In the input buffer circuit 100B of FIG. 1B, the upper-branch enabling signal ENUPB from the asymmetry upper threshold detector 114 controls the first switch 131, and the lower-branch enabling signal ENDN received from the asymmetry lower threshold detector 124 controls the second switch 132. When the upper-branch enabling signal ENUPB is at logic TRUE, the first switch 131 is at the connected state, which connects the buffer output node BufOut with the upper supply voltage VDDH. When the lower-branch enabling signal ENDN is at logic TRUE, the second switch 132 is at the connected state, which connects the buffer output node BufOut with the lower supply voltage VSS.

[0030] In the example waveform of FIG. 2F, the voltage at the buffer output node BufOut in FIG. 1B depends upon the upper-branch enabling signal ENUPB in FIG. 2D and the lower-branch enabling signal ENDN in FIG. 2E. Before time t1, because the upper-branch enabling signal ENUPB is at logic FALSE and the lower-branch enabling signal ENDN is at logic TRUE, the buffer output node BufOut is not connected to the upper supply voltage VDDH through the first switch 131 but the buffer output node BufOut is connected to the lower supply voltage VSS through the second switch 132. Consequently, the voltage at the buffer output node BufOut is at the lower supply voltage VSS.

[0031] From time t1 to time t2, because the upper-branch enabling signal ENUPB is at logic FALSE and the lower-branch enabling signal ENDN is at logic FALSE, the buffer output node BufOut is not connected to the upper supply voltage VDDH through the first switch 131 and the buffer output node BufOut is also not connected to the lower supply voltage VSS through the second switch 132. During time t1 to time t2, however, the voltage at the buffer output node BufOut is still at the lower supply voltage VSS, because the voltage at the buffer output node BufOut at time t1 is maintained until time t2 by the regenerative circuit 135 when each of the first switch 131 and the second switch 132 is not at the connecting state.

[0032] From time t2 to time t3, because the upper-branch enabling signal ENUPB is at logic TRUE and the lower-branch enabling signal ENDN is at logic FALSE, the buffer output node BufOut is connected to the upper supply voltage VDDH through the first switch 131 but the buffer output node BufOut is not connected to the lower supply voltage VSS through the second switch 132. Consequently, the voltage at the buffer output node BufOut is at the upper supply voltage VDDH.

[0033] From time t3 to time t4, because the upper-branch enabling signal ENUPB is at logic

FALSE and the lower-branch enabling signal ENDN is at logic FALSE, the buffer output node BufOut is not connected to upper supply voltage VDDH through the first switch **131** and the buffer output node BufOut is also not connected to the lower supply voltage VSS through the second switch **132**. During time **t3** to time **t4**, however, the voltage at the buffer output node BufOut is still at the upper supply voltage VDDH, because the voltage at the buffer output node BufOut at time **t3** is maintained until time **t4** by the regenerative circuit **135** when each of the first switch **131** and the second switch **132** is not at the connecting state.

[0034] After time **t4**, because the upper-branch enabling signal ENUPB is at logic FALSE and the lower-branch enabling signal ENDN is at logic TRUE, the buffer output node BufOut is not connected to the upper supply voltage VDDH through the first switch **131** rather the buffer output node BufOut is connected to the lower supply voltage VSS through the second switch **132**.

Consequently, the voltage at the buffer output node BufOut is at the lower supply voltage VSS.

[0035] In the example waveform of FIG. 2F, the voltage at the buffer output node BufOut changes from the lower supply voltage VSS to the upper supply voltage VDDH at time **t+** (which is same as time **t2**), when the input voltage signal PAD crosses the upper threshold voltage VTH as the input voltage signal PAD is rising. The voltage at the buffer output node BufOut changes from the upper supply voltage VDDH to the lower supply voltage VSS at time **t-** (which is same as time **t4**) when the input voltage signal PAD crosses a lower threshold voltage VTL as the input voltage signal PAD is falling.

[0036] An example implementation of the input buffer circuit **100B** in FIG. 1B is the input buffer circuit as shown in FIG. 1C. FIG. 1C is a schematic diagram of an input buffer circuit **100C** having the control circuit implemented with switches, in accordance with some embodiments. In the input buffer circuit **100B** of FIG. 1B, each of the asymmetry upper threshold detector **114** and the asymmetry lower threshold detector **124** can be implemented based on asymmetrical MOSFET thresholds or based on symmetrical MOSFET thresholds. In the input buffer circuit **100C** of FIG. 1C, each of the asymmetry upper threshold detector **114** and the asymmetry lower threshold detector **124** is implemented based on asymmetrical MOSFET thresholds. In FIG. 1C, each of the asymmetry upper threshold detector **114** and the asymmetry lower threshold detector **124** includes a plurality of first-type transistors serially connected with a second-type transistor.

[0037] In FIG. 1C, the asymmetry upper threshold detector **114** includes four PMOS transistors **P1u-P4u** and a NMOS transistor **N1u**, and the asymmetry lower threshold detector **124** includes four NMOS transistors **N1d-N4d** and a PMOS transistor **P1d**. The first switch **131**, which is controlled by the asymmetry upper threshold detector **114**, includes PMOS transistors **MP1** and **MP2**. The second switch **132**, which is controlled by the asymmetry lower threshold detector **124**, includes NMOS transistors **MN1** and **MN2**.

[0038] In the asymmetry upper threshold detector **114**, the channels of the PMOS transistors **P1u-P4u** are serially connected between a constant voltage node (such as the upper supply voltage VDDH) and the drain terminal of the NMOS transistor **N1u**. The drain terminal of the NMOS transistor **N1u** is the upper-branch output node **119** of the asymmetry upper threshold detector **114**. The channel of the NMOS transistor **N1u** is connected between the upper-branch output node **119** and the power supply voltages VSSH. The gate terminals of the PMOS transistors **P1u-P4u** and the NMOS transistor **N1u** are all connected together to receive the tracking-up signal PADUP from the high-side tracker **112**. The upper-branch output node **119** is configured to output an upper-branch enabling signal ENUPB which swings between two upper-branch logic levels: a logic HIGH voltage (such as the voltage VDDH) and a logic LOW voltage (such as the voltage VSSH).

[0039] In the asymmetry lower threshold detector **124**, the channels of the NMOS transistors **N1d-N4d** are serially connected between the drain terminal of the PMOS transistor **P1d** and a constant voltage node (such as the lower supply voltage VSS). The drain terminal of the PMOS transistor **P1u** is the lower-branch output node **129** of the asymmetry lower threshold detector **124**. The channel of the PMOS transistor **P1d** is connected between the power supply voltage VDDL and the

lower-branch output node **129**. The gate terminals of the NMOS transistors **N1d-N4d** and the PMOS transistor **P1d** are all connected together to receive tracking-down signal **PADDN** from the low-side tracker **122**. The lower-branch output node **129** of the asymmetry lower threshold detector **124** is configured to output a lower-branch enabling signal **ENDN** which swings between two lower-branch logic levels: a logic HIGH voltage (such as the voltage **VDDL**) and a logic LOW voltage (such as the voltage **VSS**).

[0040] In the first switch **131**, the channels of the PMOS transistors **MP1** and **MP2** are serially connected between the upper supply voltage **VDDH** and the buffer output node **BufOut**. The gate terminal of the PMOS transistor **MP1** is connected to the upper-branch output node **119** of the asymmetry upper threshold detector **114** to receive the upper-branch enabling signal **ENUPB**. The gate terminal of the PMOS transistor **MP1** is connected to the power supply voltages **VSSH**.

[0041] In the second switch **132**, the channels of the NMOS transistors **MN2** and **MN1** are serially connected between the buffer output node **BufOut** and the lower supply voltage **VSS**. The gate terminal of the NMOS transistor **MN1** is connected to the lower-branch output node **129** of the asymmetry lower threshold detector **124** to receive the lower-branch enabling signal **ENDN**. The gate terminal of the NMOS transistor **MN1** is connected to the power supply voltage **VDDL**.

[0042] In operation, if an input voltage signal **PAD** as shown in FIG. 2A is coupled to the input node **102** of the input buffer circuit **100C** in FIG. 1C, the voltage at the buffer output node **BufOut** of the input buffer circuit **100C** has the waveform as shown in FIG. 2F.

[0043] Before time **t1**, the tracking-up signal **PADUP** at the output of the high-side tracker **112** as induced by the input voltage signal **PAD** is below the upper threshold voltage **VTH** of the asymmetry upper threshold detector **114**, and the logic HIGH voltage (such as the voltage **VDDH**) at the output of the asymmetry upper threshold detector **114** drives the PMOS transistor **MP1** into the non-conducting state. As the PMOS transistor **MP1** is driven into the non-conducting state, the first switch **131** is driven into the disconnected state. Here, the logic HIGH voltage (such as the voltage **VDDH**) of the upper-branch enabling signal **ENUPB** at the output of the asymmetry upper threshold detector **114** is logic FALSE for the purpose of controlling the first switch **131**.

Additionally, before time **t1**, the tracking-down signal **PADDN** at the output of the low-side tracker **122** as induced by the input voltage signal **PAD** is below the lower threshold voltage **VTL** of the asymmetry lower threshold detector **124**, and the logic HIGH voltage (such as the voltage **VDDL**) at the output of the asymmetry lower threshold detector **124** drives the NMOS transistor **MN1** into the conducting state. In response to both the NMOS transistors **MN1** and **MN2** being at the conducting state, the second switch **132** is driven into the connected state. Here, the logic HIGH voltage (such as the voltage **VDDL**) of the lower-branch enabling signal **ENDN** at the output of the asymmetry lower threshold detector **124** is logic TRUE for the purpose of controlling the second switch **132**.

[0044] Before time **t1**, as shown in FIG. 2F, the buffer output node **BufOut** is at voltage **VSS**, in response to the condition that the first switch **131** is at the disconnected state and the second switch **132** is at the connected state.

[0045] From time **t1** to time **t2**, the tracking-up signal **PADUP** is still below the upper threshold voltage **VTH** of the asymmetry upper threshold detector **114**, and the output of the asymmetry upper threshold detector **114** is still at the logic HIGH voltage (such as the voltage **VDDH**) which drives the PMOS transistor **MP1** into the non-conducting state. Consequently, the first switch **131** is still at the disconnected state, while the PMOS transistor **MP1** is maintained at the non-conducting state. On the other hand, as the tracking-down signal **PADDN** rises above the lower threshold voltage **VTL** of the asymmetry lower threshold detector **124** at time **t1**, the output of the asymmetry lower threshold detector **124** starts to change from the logic HIGH voltage (such as the voltage **VDDL**) towards the logic LOW voltage (such as the voltage **VSS**). In response to the output of the asymmetry lower threshold detector **124** reaching the logic LOW voltage (such as the voltage **VSS**), the NMOS transistor **MN1** is driven into the non-conducting state, and consequently

the second switch **132** is driven into the disconnected state. Here, the logic LOW voltage (such as the voltage VSS) of the lower-branch enabling signal ENDN at the output of the asymmetry lower threshold detector **124** is logic FALSE for the purpose of controlling the second switch **132**.

[0046] From time t_1 to time t_2 , as shown in FIG. 2F, the buffer output node BufOut is still at voltage VSS, even the first switch **131** is at the disconnected state and the second switch **132** is changed from the connected state to the disconnected state after some delay from time t_1 . The voltage at buffer output node BufOut is maintained by the regenerative circuit **135**, even both the first switch **131** and the second switch **132** are at the disconnected state before time t_2 .

[0047] At time t_2 , the tracking-up signal PADUP rises above the upper threshold voltage VTH of the asymmetry upper threshold detector **114**, and the output of the asymmetry upper threshold detector **114** starts to change from the logic HIGH voltage (such as the voltage VDDH) towards the logic LOW voltage (such as the voltage VSSH). In response to the output of the asymmetry upper threshold detector **114** reaching the logic LOW voltage (such as the voltage VSSH), the PMOS transistor MP1 is driven into the conducting state, and consequently the first switch **131** is driven into the connected state. Here, the logic LOW voltage (such as the voltage VSSH) of the upper-branch enabling signal ENUPB at the output of the asymmetry upper threshold detector **114** is logic TRUE for the purpose of controlling the first switch **131**. Additionally, at time t_2 and thereafter, the tracking-down signal PADDN continues to be above the lower threshold voltage VTL of the asymmetry lower threshold detector **124** at least until time t_4 . In proper operation, the output of the asymmetry lower threshold detector **124** reaches the logic LOW voltage (such as the voltage VSS) before time t_2 . At time t_2 and thereafter, the output of the asymmetry lower threshold detector **124** is maintained at the logic LOW voltage (such as the voltage VSS) at least until time t_4 , and consequently the second switch **132** is maintained at the disconnected state at least until time t_4 . Again, the logic LOW voltage (such as the voltage VSS) of the lower-branch enabling signal ENDN at the output of the asymmetry lower threshold detector **124** is logic FALSE for the purpose of controlling the second switch **132**.

[0048] After some delay from time t_2 , in response to the output of the asymmetry upper threshold detector **114** reaching the logic LOW voltage (such as the voltage VSSH), the first switch **131** is driven into the connected state, and consequently the voltage at the buffer output node BufOut is changed to the voltage VDDH, while the second switch **132** is maintained at the disconnected state.

[0049] From time t_2 to time t_3 , as shown in FIG. 2F, the buffer output node BufOut is at the voltage VDDH, in response to the condition that the first switch **131** is at the connected state after some delay from time t_2 and the second switch **132** is at the disconnected state since time t_2 .

[0050] At time t_3 , the tracking-up signal PADUP falls below the upper threshold voltage VTH of the asymmetry upper threshold detector **114**, and the output of the asymmetry upper threshold detector **114** starts to change from the logic LOW voltage (such as the voltage VSSH) towards the logic HIGH voltage (such as the voltage VDDH). After some delay from time t_3 , in response to the output of the asymmetry upper threshold detector **114** reaching the logic HIGH voltage (such as the voltage VDDH), the PMOS transistor MP1 is driven into the non-conducting state, and consequently the first switch **131** is driven into the disconnected state. Additionally, at time t_3 and thereafter, the tracking-down signal PADDN continues to be above the lower threshold voltage VTL of the asymmetry lower threshold detector **124** at least until time t_4 , and the output of the asymmetry lower threshold detector **124** is maintained at the logic LOW voltage (such as the voltage VSS) at least until time t_4 . Consequently, the second switch **132** is maintained at the disconnected state from time t_3 to at least time t_4 .

[0051] From time t_3 to time t_4 , as shown in FIG. 2F, the buffer output node BufOut is maintained at voltage VDDH, even the first switch **131** is changed from the connected state to the disconnected state after some delay from time t_3 and the second switch **132** is at the disconnected state before time t_4 . The voltage at buffer output node BufOut is maintained by the regenerative circuit **135**, even both the first switch **131** and the second switch **132** are at the disconnected state before time

t4.

[0052] At time t4 and thereafter, the tracking-up signal PADUP continues to be below the upper threshold voltage VTH of the asymmetry upper threshold detector 114. In proper operation, the output of the asymmetry upper threshold detector 114 reaches the logic HIGH voltage (such as the voltage VDDH) before time t4 and is maintained at the logic HIGH voltage (such as the voltage VDDH) thereafter. Consequently, the output of the asymmetry upper threshold detector 114, the PMOS transistor MP1 is maintained at the non-conducting state, and the first switch 131 is maintained at the disconnected state after time t4. Additionally, at time t4, the tracking-down signal PADDN falls below the lower threshold voltage VTL of the asymmetry lower threshold detector 124, and the output of the asymmetry lower threshold detector 124 starts to change from the logic LOW voltage (such as the voltage VSS) towards the logic HIGH voltage (such as the voltage VDDL). In response to the output of the asymmetry lower threshold detector 124 reaching the logic HIGH voltage (such as the voltage VDDL), the NMOS transistor MN1 is driven into the conducting state after some delay from time t4, and consequently the second switch 132 is driven into the connected state.

[0053] After time t4, as shown in FIG. 2F, the voltage at the buffer output node BufOut is changed to voltage VSS, in response to the condition that the first switch 131 is maintained at the disconnected state since time t4 and the second switch 132 is changed from the disconnected state to the connected state after some delay from time t4.

[0054] In FIG. 2D, the example waveform of the upper-branch enabling signal ENUPB is generated with an asymmetry upper threshold detector 114 in which the PMOS transistors P1u-P4u and the NMOS transistor N1u are sufficiently fast as compared with the falling edge of the input voltage signal PAD in FIG. 2A. Specifically, the PMOS transistors P1u-P4u and the NMOS transistor N1u are sufficiently fast such that the output of the asymmetry upper threshold detector 114 reaches the logic HIGH voltage (such as the voltage VDDH) before time t4, after the input voltage signal PAD falls below the upper threshold voltage VTH of the asymmetry upper threshold detector 114 at time t3 when the output of the asymmetry upper threshold detector 114 starts to change from the logic LOW voltage (such as the voltage VSSH) towards the logic HIGH voltage (such as the voltage VDDH). In some circumstances, as shown in the example waveform of FIG. 3D, because the transistors in the asymmetry upper threshold detector 114 are too slow, the output of the asymmetry upper threshold detector 114 is unable to reach the logic HIGH voltage (such as the voltage VDDH) at time t4, which causes an undesirable situation such that both the first switch 131 and the second switch 132 are at the connected state.

[0055] In FIG. 2E, the example waveform of the lower-branch enabling signal ENDN is generated with an asymmetry lower threshold detector 124 in which the NMOS transistors N1u-N4u and the PMOS transistor P1d are sufficiently fast as compared with the rising edge of the input voltage signal PAD in FIG. 2A. Specifically, the NMOS transistors N1u-N4u and the PMOS transistor P1d are sufficiently fast such that the output of the asymmetry lower threshold detector 124 reaches towards the logic LOW voltage (such as the voltage VSS) before time t2, after the tracking-down signal PADDN rises above the lower threshold voltage VTL of the asymmetry lower threshold detector 124 at time t1 when the output of the asymmetry lower threshold detector 124 starts to change from the logic HIGH voltage (such as the voltage VDDL) towards the logic LOW voltage (such as the voltage VSS). In some circumstances, as shown in the example waveform of FIG. 3E, because the transistors in the asymmetry lower threshold detector 124 are too slow, the output of the asymmetry lower threshold detector 124 is unable to reach the logic LOW voltage (such as the voltage VSS) at time t2, which causes an undesirable situation such that both the first switch 131 and the second switch 132 are at the connected state.

[0056] FIGS. 3A-3F are diagrams of waveforms of signals at the input node, the output node, and various other nodes in the input buffer circuit 100C in FIG. 1C, in accordance with some embodiments. FIGS. 3A-3C are correspondingly the same as FIGS. 2A-2C.

[0057] FIG. 3D is the waveform of the upper-branch enabling signal ENUPB at the output of the asymmetry upper threshold detector **114** in the input buffer circuit **100C** of FIG. 1C. Because the asymmetry upper threshold detector **114** in the input buffer circuit **100C** is an asymmetrical threshold detector based on asymmetrical MOSFET thresholds, the falling time of the upper-branch enabling signal ENUPB starting at time t_2 to change from the logic HIGH voltage to the logic LOW voltage is smaller than the rising time of the upper-branch enabling signal ENUPB starting at time t_3 to change from the logic LOW voltage to HIGH voltage. In some circumstances, as shown in FIG. 3D, the upper-branch enabling signal ENUPB reaches the logic HIGH voltage at time t_3^* , and the time t_3^* is later than time t_4 (which is the instant the input voltage signal PAD falls below the lower threshold voltage VTL). The slower the transistors in the asymmetry upper threshold detector **114**, the larger the time difference $t_3^*-t_3$. The faster the input voltage signal PAD decreases at the falling edge from the time t_3 to time t_4 , the smaller the time difference t_4-t_3 . In some circumstances, due to a combination of slower transistors and fast signal swings of the input voltage signal PAD, the time difference $t_3^*-t_3$ is larger than the time difference t_4-t_3 , and consequently the time t_3^* is later than the time t_4 .

[0058] During the time period from the time t_4 to the time t_3^* , the upper-branch enabling signal ENUPB is at logic TRUE (at the logic LOW voltage) which causes the first switch **131** to be at the connected state, while the lower-branch enabling signal ENDN is also at logic TRUE (at the logic HIGH voltage) which causes the second switch **132** to be at the connected state. In response to the condition that both the first switch **131** and the second switch **132** are at the connected state between the time t_4 and the time t_3^* , malfunctions in the asymmetry upper threshold detector **114** may be induced by a short circuit.

[0059] FIG. 3E is the waveform of the lower-branch enabling signal ENDN at the output of the asymmetry lower threshold detector **124** in the input buffer circuit **100C** of FIG. 1C. Because the asymmetry lower threshold detector **124** in the input buffer circuit **100C** is an asymmetrical threshold detector based on asymmetrical MOSFET thresholds, the falling time of the lower-branch enabling signal ENDN starting at time t_1 to change from the logic HIGH voltage to the logic LOW voltage is larger than the rising time of the lower-branch enabling signal ENDN starting at time t_4 to change from the logic LOW voltage to the logic HIGH voltage. In some circumstances, as shown in FIG. 3E, the lower-branch enabling signal ENDN reaches the logic LOW voltage at time t_1^* which is later than time t_2 (which is the instant the input voltage signal PAD rises above the upper threshold voltage VTH). The slower the transistors in the asymmetry lower threshold detector **124**, the larger the time difference $t_1^*-t_1$. The faster the input voltage signal PAD increases at the rising edge from the time t_1 to time t_2 , the smaller the time difference t_2-t_1 . In some circumstances, due to a combination of slower transistors and fast signal swings of the input voltage signal PAD, the time difference $t_1^*-t_1$ is larger than the time difference t_2-t_1 , and consequently the time t_1^* is later than the time t_2 .

[0060] During the time period from the time t_2 to the time t_1^* , the lower-branch enabling signal ENDN is at logic TRUE (at the logic HIGH voltage) which causes the second switch **132** to be at the connected state, while the upper-branch enabling signal ENUPB is also at logic TRUE (at the logic LOW voltage) which causes the first switch **131** to be at the connected state. In response to the condition that both the first switch **131** and the second switch **132** are at the connected state between the time t_2 and the time t_1^* , malfunctions in the asymmetry lower threshold detector **124** may be induced by a short circuit.

[0061] FIG. 3F is the waveform of the voltage at the buffer output node BufOut in the input buffer circuit **100C** of FIG. 1C. In FIG. 3E, the asymmetry lower threshold detector **124** risks malfunctions from time t_+ to the time t_1^* , and the asymmetry upper threshold detector **114** risks malfunctions from the time t_- and the time t_3^* . The risks of malfunctions in the asymmetry upper threshold detector **114** and in the asymmetry lower threshold detector **124** due to fast signal swings of the input voltage signal PAD are reduced with improved input buffer circuits, such as the input

buffer circuits **400A-400B** in FIGS. **4A-4B**.

[0062] An integrated circuit often has multiple instances of the input buffer circuits, and consequentially an integrated circuit often has multiple instances of the asymmetry upper threshold detector and multiple instances of lower threshold detectors. Because of process, voltage, and temperature variations ("PVT variations") and changes in reliability degradation tolerances, the time $t3^*$ associated with each instance of the asymmetry upper threshold detector **114** and the time $t1^*$ associated with each instance of the asymmetry lower threshold detector **124** all have variations according to some statistical distributions. The statistical distribution of the time $t3^*$ corresponds to a time spread of the time $t3^*$, which is depicted as multiple traces between time $t3$ and $t3^*$ in FIG. **3D**. The statistical distribution of the time $t1^*$ corresponds to a time spread of the time $t1^*$, which is depicted as multiple traces between time $t1$ and $t1^*$ in FIG. **3E**.

[0063] The larger PVT variations, the larger the time spread of the time $t3^*$. The larger the time spread of the time $t3^*$, the larger the number of instances of the asymmetry upper threshold detector **114** that risks malfunctions because of the undesirable positive time delay $t3^*-t4>0$. The risk of malfunctions between time $t4$ and time $t3^*$ also increases as the operating frequency of the integrated circuit increases. Similarly, The larger PVT variations, the larger the time spread of the time $t1^*$. The larger the time spread of the time $t1^*$, the larger the number of instances of the asymmetry lower threshold detector **124** that risks malfunctions because of the undesirable positive time delay $t1^*-t2>0$. The risk of malfunctions between time $t2$ and time $t1^*$ also increases as the operating frequency of the integrated circuit increases. For high-speed operations of an integrated circuit which has multiple instances of the input buffer circuits, the input buffer circuits **400A-400B** in FIGS. **4A-4B** provide improved reliability and reduced malfunctions as compared with the input buffer circuit **100C** in FIG. **1C**.

[0064] FIGS. **4A-4B** are schematic diagrams of input buffer circuits having variable thresholds in threshold circuits, in accordance with some embodiments. Each of the input buffer circuits **400A-400B** in FIGS. **4A-4B** includes an asymmetry upper threshold circuit **410** and an asymmetry lower threshold circuit **420** both configured to receive a threshold-selection signal **450**. Based on the threshold-selection signal **450** received by the asymmetry upper threshold circuit **410**, the asymmetry upper threshold circuit **410** is configured to operate either as the asymmetry upper threshold detector **114** or as a supplementary upper threshold detector **116**. In response to the threshold-selection signal **450** indicating that the input voltage signal PAD increases at the rising edge, the asymmetry upper threshold circuit **410** operates as the asymmetry upper threshold detector **114** with the main upper threshold VTH. Conversely, In response to the threshold-selection signal **450** indicating that the input voltage signal PAD decreases at the falling edge, the asymmetry upper threshold circuit **410** operates as the supplementary upper threshold detector **116** with the supplementary upper threshold VTH+. The supplementary upper threshold VTH+ is higher than the main upper threshold VTH.

[0065] Based on the threshold-selection signal **450** received by the asymmetry lower threshold circuit **420**, the asymmetry lower threshold circuit **420** is configured to operate either as the asymmetry lower threshold detector **124** or as a supplementary lower threshold detector **126**. In response to the threshold-selection signal **450** indicating that the input voltage signal PAD decreases at the falling edge, the asymmetry lower threshold circuit **420** operates as the asymmetry lower threshold detector **124** with the main lower threshold VTL. Conversely, In response to the threshold-selection signal **450** indicating that the input voltage signal PAD increases at the rising edge, the asymmetry lower threshold circuit **420** operates as the supplementary lower threshold detector **126** with the supplementary lower threshold VTL-. The supplementary lower threshold VTL- is lower than the main lower threshold VTL.

[0066] FIGS. **5A-5F** are diagrams of waveforms of signals at the input node, the output node, and various other nodes in an improved input buffer circuit (such as the input buffer circuit **400A** or **400B** in FIGS. **4A-4B**), in accordance with some embodiments. FIG. **5A** is the waveform of the

input voltage signal PAD at the input node **102**. FIG. 5B is the waveform of the tracking-up signal PADUP at the output of the high-side tracker **112**. FIG. 5C is the waveform of the tracking-down signal PADDN at the output of the low-side tracker **122**. FIG. 5D is the waveform of the upper-branch enabling signal ENUPB at the output of the asymmetry upper threshold circuit **410**. FIG. 5E is the waveform of the lower-branch enabling signal ENDN at the output of the asymmetry lower threshold circuit **420**. FIG. 5F is the waveform of the voltage at the buffer output node BufOut. While FIG. 5A and FIG. 5F are correspondingly the same as FIG. 2A and FIG. 2F, FIGS. 5B-5E are correspondingly modified from FIGS. 3B-3E.

[0067] As the input voltage signal PAD in FIG. 5A increases at the rising edge from time t_a to time t_b , the asymmetry upper threshold circuit **410** operates as the asymmetry upper threshold detector **114** with the main upper threshold V_{TH} , and the asymmetry lower threshold circuit **420** operates as the supplementary lower threshold detector **126** with the supplementary lower threshold V_{TL-} . In response to the tracking-down signal PADDN in FIG. 5C rising across the supplementary lower threshold V_{TL-} before t_1 , the lower-branch enabling signal ENDN (in FIG. 5E) at the output of the asymmetry lower threshold circuit **420** starts to change from the logic HIGH voltage towards the logic LOW voltage. The lower-branch enabling signal ENDN reaches the logic LOW voltage before time t_2 . Here, time t_2 is the instant when the tracking-up signal PADUP in FIG. 5B rises across the main upper threshold V_{TH} . That is, time t_2 is the instant when the upper-branch enabling signal ENUPB (in FIG. 5D) at the output of the asymmetry upper threshold circuit **410** starts to change from the logic HIGH voltage towards the logic LOW voltage.

[0068] In FIG. 5E, after the tracking-down signal PADDN (in FIG. 5C) rises across the supplementary lower threshold V_{TL-} before time t_1 , the lower-branch enabling signal ENDN reaches the logic LOW voltage before time t_2 , because the asymmetry lower threshold circuit **420** operates as the supplementary lower threshold detector **126** with the supplementary lower threshold V_{TL-} . As a comparison, in FIG. 3E, after the tracking-down signal PADDN (in FIG. 3C) rises across the lower threshold V_{TL} at time t_1 , the lower-branch enabling signal ENDN in the input buffer circuit **100C** of FIG. 1C reaches the logic LOW voltage at time t_1^* which is later than time t_2 . Unlike the input buffer circuit **100C** of FIG. 1C which has the risk of circuit malfunctions between time t_2 and time t_1^* , the improved input buffer circuit (such as the input buffer circuit **400A** or **400B** in FIGS. 5A-4B) reduces the risk of circuit malfunctions after time t_2 , because the lower-branch enabling signal ENDN reaches the logic LOW voltage before time t_2 . The time spread of the time t_1^* after time t_2 is suppressed, as shown by the arrow pointing towards the TRUE-to-FALSE falling edge near time t_1 in FIG. 5E.

[0069] As the input voltage signal PAD in FIG. 5A decreases at the falling edge from time t_c to time t_d , the asymmetry upper threshold circuit **410** operates as the supplementary upper threshold detector **116** with the supplementary upper threshold V_{TH+} , and the asymmetry lower threshold circuit **420** operates as the asymmetry lower threshold detector **124** with the main lower threshold V_{TL} . In response to the tracking-up signal PADUP in FIG. 5B falling across the supplementary upper threshold V_{TH+} , the upper-branch enabling signal ENUPB (in FIG. 5D) at the output of the asymmetry upper threshold circuit **410** starts to change from the logic LOW voltage towards the logic HIGH voltage. The upper-branch enabling signal ENUPB reaches the logic HIGH voltage before time t_4 . Here, time t_4 is the instant when the tracking-down signal PADDN in FIG. 5C falls across the main lower threshold V_{TL} . That is, time t_4 is the instant when the lower-branch enabling signal ENDN (in FIG. 5E) at the output of the asymmetry lower threshold circuit **420** starts to change from the logic LOW voltage towards the logic HIGH voltage.

[0070] In FIG. 5D, after the tracking-up signal PADUP in FIG. 5B falls across the supplementary upper threshold V_{TH+} before time t_3 , the upper-branch enabling signal ENUPB (in FIG. 5D) reaches the logic HIGH voltage before time t_4 , because the asymmetry upper threshold circuit **410** operates as the supplementary upper threshold detector **116** with the supplementary upper threshold V_{TH+} . As a comparison, in FIG. 3D, after the tracking-up signal PADUP in FIG. 3B falls across

the upper threshold VTH at time t3, the upper-branch enabling signal ENUPB in the input buffer circuit **100C** of FIG. **1C** reaches the logic HIGH voltage at time t3* which is later than time t4. Unlike the input buffer circuit **100C** of FIG. **1C** which has the risk of circuit malfunctions between time t4 and time t3*, the improved input buffer circuit (such as the input buffer circuit **400A** or **400B** in FIGS. **4A-4B**) removed the risk of circuit malfunctions after time t4, because the upper-branch enabling signal ENUPB reaches the logic HIGH voltage before time t4. The time spread of the time t3* after time t4 is suppressed, as shown by the arrow pointing towards the TRUE-to-FALSE rising edge near time t3 in FIG. **5D**.

[0071] The input buffer circuit **400B** in FIG. **4B**, which provides an example implementation of the input buffer circuit **400A** in FIG. **4A**, is modified from the input buffer circuit **100C** of FIG. **1C**. Based on the input buffer circuit **100C** of FIG. **1C**, a transmission gate **118** and a transmission gate **128** are added as the elements of the input buffer circuit **400B**.

[0072] In FIG. **4B**, the input buffer circuit **400B** includes an asymmetry upper threshold circuit **410** and an asymmetry lower threshold circuit **420**. The asymmetry upper threshold circuit **410** in FIG. **4B**, which is modified from the asymmetry upper threshold detector **114** in FIG. **1C**, includes a transmission gate **118** connected between the drain terminal of the PMOS transistor P1u and the drain terminal of the NMOS transistor N1u. The asymmetry lower threshold circuit **420** in FIG. **4B**, which is modified from the asymmetry lower threshold detector **124** in FIG. **1C**, includes a transmission gate **128** connected between the drain terminal of the PMOS transistor P1d and the drain terminal of the NMOS transistor N4d.

[0073] In operation, each of the transmission gate **118** and the transmission gate **128** is controlled with the threshold-selection signal **450**. The operation mode of the asymmetry upper threshold circuit **410** and the operation mode of the asymmetry lower threshold circuit **420** depend upon the threshold-selection signal **450** received.

[0074] In response to the threshold-selection signal **450** indicating that the input voltage signal PAD increases at the rising edge, the transmission gate **118** is set to the non-transmission state which blocks the direct conductive connection between the drain terminal of the PMOS transistor P1u and the drain terminal of the NMOS transistor N1u, and the transmission gate **128** is set to the transmission state which forms the direct conductive connection between the drain terminal of the PMOS transistor P1d and the drain terminal of the NMOS transistor N4d. Consequently, at the rising edge of the input voltage signal PAD, the asymmetry upper threshold circuit **410** operates as the asymmetry upper threshold detector **114** with the main upper threshold VTH, and the asymmetry lower threshold circuit **420** operates as the supplementary lower threshold detector **126** with the supplementary lower threshold VTL-.

[0075] In response to the threshold-selection signal **450** indicating that the input voltage signal PAD decreases at the falling edge, the transmission gate **118** is set to the transmission state which forms the direct conductive connection between the drain terminal of the PMOS transistor P1u and the drain terminal of the NMOS transistor N1u, and the transmission gate **128** is set to the non-transmission state which blocks the direct conductive connection between the drain terminal of the PMOS transistor P1d and the drain terminal of the NMOS transistor N4d. Consequently, at the falling edge of the input voltage signal PAD, the asymmetry upper threshold circuit **410** operates as the supplementary upper threshold detector **116** with the supplementary upper threshold VTH+, and the asymmetry lower threshold circuit **420** operates as the asymmetry lower threshold detector **124** with the main lower threshold VTL.

[0076] In the input buffer circuit **400B** in FIG. **4B**, the transmission gate **118** and the transmission gate **128** are both controlled with the threshold-selection signal **450**. In some alternative embodiments, the transmission gate **118** and the transmission gate **128** are controlled with two separate and related threshold-selection signals. In some embodiments, threshold-selection signal **450** or other related threshold-selection signals are generated based on the voltage signal at the buffer output node BufOut of the input buffer circuit. One example method of generating threshold-

selection signals is disclosed with respect to FIG. 6.

[0077] FIG. 6 is a schematic diagram of an input buffer circuit **600** which has threshold circuits controlled with threshold-selection signals, in accordance with some embodiments. The input buffer circuit **600** in FIG. 6 is modified from the input buffer circuit **400B** in FIG. 4B. In FIG. 6, threshold-selection signals **450U** and **450D** replace the threshold-selection signal **450** in FIG. 4B and correspondingly control the transmission gates **118** and **128**. Level shifters **140U** and **140D** replace the level shifter **140** in FIG. 4B. Delay circuits **610U** and **610D** and an inverter **620** are added. The delay circuit **610U** is coupled between the level shifter **140U** and the transmission gate **118**. The delay circuit **610D** and the inverter **620** are coupled between the level shifter **140D** and the transmission gate **128**.

[0078] In operation, the controlling of the transmission gates **118** and **128** is based on the voltage signal at the buffer output node BufOut of the input buffer circuit. As shown in FIG. 5A and FIG. 5F, before the input voltage signal PAD increases at the rising edge, the voltage signal at the buffer output node BufOut is at logic LOW. The threshold-selection signal **450U** is set to logic FALSE, which is induced by the voltage signal at the buffer output node BufOut through the delay circuit **610U**. The threshold-selection signal **450D** is set to logic TRUE, which is induced by the voltage signal at the buffer output node BufOut through the delay circuit **610D** and the inverter **620**. In response to the threshold-selection signal **450U** being at logic FALSE, the transmission gate **118** is set to the non-transmission state. In response to the threshold-selection signal **450D** being at logic TRUE, the transmission gate **128** is set to the transmission state.

[0079] Furthermore, as shown in FIG. 5A and FIG. 5F, before the input voltage signal PAD decreases at the falling edge, the voltage signal at the buffer output node BufOut is at logic HIGH. The threshold-selection signal **450U** is set to logic TRUE, which is induced by the voltage signal at the buffer output node BufOut through the delay circuit **610U**. The threshold-selection signal **450D** is set to logic FALSE, which is induced by the voltage signal at the buffer output node BufOut through the delay circuit **610D** and the inverter **620**. In response to the threshold-selection signal **450U** being at logic TRUE, the transmission gate **118** is set to the transmission state. In response to the threshold-selection signal **450D** being at logic FALSE, the transmission gate **128** is set to the non-transmission state. In FIG. 6, the threshold-selection signal **450U** is generated by a first threshold-selection signal circuit having the delay circuit **610U**, and the threshold-selection signal **450D** is generated by a second threshold-selection signal circuit having the delay circuit **610D** and the inverter **620**.

[0080] In the embodiments of FIG. 4B and FIG. 6, each of the transmission gates **118** and **128** functions as a liner switch. In some alternative embodiments, either one of the transmission gates **118** and **128** or both of the transmission gates **118** and **128** are replaced with other forms of linear switches.

[0081] FIG. 7 is a flowchart of method **700** of generating an output voltage signal from an input voltage signal, in accordance with some embodiments. The sequence in which the operations of method **700** are depicted in FIG. 7 is for illustration only; the operations of method **700** are capable of being executed in sequences that differ from that depicted in FIG. 7. It is understood that additional operations may be performed before, during, and/or after the method **700** depicted in FIG. 7, and that some other processes may only be briefly described herein. The operations of the method **700** are described by referring to the waveforms in FIG. 5A-5F and the input buffer circuit **400B** in FIG. 4B.

[0082] In operation **705** of method **700**, the input buffer circuit determines whether the input voltage signal PAD is rising or falling. If the input voltage signal PAD is rising, the process proceeds to operation **710**. If the input voltage signal PAD is falling, the process proceeds to operation **720**.

[0083] In operation **710**, in response to the input voltage signal PAD rising across a supplementary lower threshold VTL-(as shown in FIG. 5C), the input buffer circuit changes the lower-branch

enabling signal from logic TRUE to logic FALSE (as shown in FIG. 5E). In the embodiments of FIG. 4B, for the lower-branch enabling signal ENDN, logic TRUE is represented with the logic HIGH voltage (such as the voltage VDDL), and logic FALSE is represented with the logic LOW voltage (such as the voltage VSS).

[0084] Then, in operation **714** following operation **710**, the second switch **132** is driven into the disconnected state. In the embodiments of FIG. 4B, the logic LOW voltage (such as the voltage VSS) is applied to the gate terminal of the NMOS transistor MN1, which drives the second switch **132** into the disconnected state.

[0085] In operation **730**, in response to the input voltage signal PAD rising across a main upper threshold VTH (as shown in FIG. 5B), the input buffer circuit changes the upper-branch enabling signal from logic FALSE to logic TRUE (as shown in FIG. 5D). In the embodiments of FIG. 4B, for the upper-branch enabling signal ENUPB, logic TRUE is represented with the logic LOW voltage (such as the voltage VSSH), and logic FALSE is represented with the logic HIGH voltage (such as the voltage VDDH).

[0086] Then, in operation **734** following operation **730**, the first switch **131** is driven into the connected state. In the embodiments of FIG. 4B, the logic LOW voltage (such as the voltage VSSH) is applied to the gate terminal of the PMOS transistor MP1, which drives the first switch **131** into the connected state. After operation **734**, the process proceeds to operation **705**.

[0087] In the event that, in operation **705**, the input buffer circuit determines that the input voltage signal PAD is falling, then, the process proceeds to operation **720**. In operation **720**, in response to the input voltage signal PAD falling across a supplementary upper threshold VTH+ (as shown in FIG. 5B), the input buffer circuit changes the upper-branch enabling signal from logic TRUE to logic FALSE (as shown in FIG. 5D). In the embodiments of FIG. 4B, for the upper-branch enabling signal ENUPB, logic TRUE is represented with the logic LOW voltage (such as the voltage VSSH), and logic FALSE is represented with the logic HIGH voltage (such as the voltage VDDH).

[0088] Then, in operation **724** following operation **720**, the first switch **131** is driven into the disconnected state. In the embodiments of FIG. 4B, the logic HIGH voltage (such as the voltage VDDH) is applied to the gate terminal of the PMOS transistor MP1, which drives the first switch **131** into the disconnected state.

[0089] In operation **740**, in response to the input voltage signal PAD falling across a main lower threshold VTL (as shown in FIG. 5C), the input buffer circuit changes the lower-branch enabling signal from logic FALSE to logic TRUE (as shown in FIG. 5E). In the embodiments of FIG. 4B, for the lower-branch enabling signal ENDN, logic TRUE is represented with the logic HIGH voltage (such as the voltage VDDL), and logic FALSE is represented with the logic LOW voltage (such as the voltage VSS).

[0090] Then, in operation **744** following operation **740**, the second switch **132** is driven into the connected state. In the embodiments of FIG. 4B, the logic HIGH voltage (such as the voltage VDDL) is applied to the gate terminal of the NMOS transistor MN1, which drives the second switch **132** into the connected state.

[0091] In the example embodiments as shown in FIG. 1C, FIG. 4B, and FIG. 6, the asymmetry upper threshold detector **114** includes four PMOS transistors P1u-P4u stacked together and a NMOS transistor N1u, and the asymmetry lower threshold detector **124** includes four NMOS transistors N1d-N4d stacked together and a PMOS transistor P1d. In some alternative embodiments, the asymmetry upper threshold detector **114** includes less than four PMOS transistors or more than four PMOS transistors. In some alternative embodiments, the asymmetry lower threshold detector **124** includes less than four NMOS transistors or more than four NMOS transistors. Implementations of the asymmetry upper threshold detector **114** with multiple stacked PMOS transistors other than stacked four PMOS transistors are within the contemplated scope of present disclosure. Implementations of the asymmetry lower threshold detector **124** with multiple stacked NMOS transistors other than four stacked NMOS transistors are also within the

contemplated scope of present disclosure.

[0092] In some embodiments, in the asymmetry upper threshold detector **114**, each of the four PMOS transistors **P1u-P4u** is a standard threshold transistor. In some alternative embodiments, the PMOS transistor **P1u** is a low threshold transistor, while the remaining three PMOS transistors **P2u-P4u** are standard threshold transistors. In still some alternative embodiments, the four stacked PMOS transistors **P1u-P4u** in the asymmetry upper threshold detector **114** are substituted with multiple stacked PMOS transistors **P1u**, **P2u**, . . . , and **Pku**, where **k** is an integer that is larger than two. Among the multiple stacked PMOS transistors, the PMOS transistor **P1u** is a low threshold transistor, while the remaining (**k-1**) PMOS transistors **P2u-Pku** are standard threshold transistors. As the PMOS transistor **P1u** is changed from a standard threshold transistor to a low threshold transistor, the upper-branch enabling signal **ENUPB** at the output of the asymmetry upper threshold detector **114** has an improved transition speed from logic TRUE to logic FALSE. Some other embodiments of the asymmetry upper threshold detector **114** are described with respect to FIG. **8A**.

[0093] In some embodiments, in the asymmetry lower threshold detector **124**, each of the four NMOS transistors **N1d-N4d** is a standard threshold transistor. In some alternative embodiments, the NMOS transistor **N4d** is a low threshold transistor, while the remaining three NMOS transistors **N1d-N3d** are standard threshold transistors. In still some alternative embodiments, the four stacked NMOS transistors **N1d-N4d** in the asymmetry lower threshold detector **124** are substituted with multiple stacked NMOS transistors **N1d**, **N2d**, . . . , and **Nkd**, where **k** is an integer that is larger than two. Among the multiple stacked NMOS transistors, the NMOS transistor **Nkd** is a low threshold transistor, while the remaining (**k-1**) NMOS transistors **N1d**, **N2d**, . . . , and **N(k-1)d** are standard threshold transistors. As the NMOS transistor **Nkd** is changed from a standard threshold transistor to a low threshold transistor, the lower-branch enabling signal **ENDN** at the output of the asymmetry lower threshold detector **124** has an improved transition speed from logic TRUE to logic FALSE. Some other embodiments of the asymmetry lower threshold detector **124** are described with respect to FIG. **8B**.

[0094] FIG. **8A** is a schematic diagram of an asymmetry upper threshold circuit, in accordance with some embodiments. The asymmetry upper threshold detector in the asymmetry upper threshold circuit **810** of FIG. **8A** is modified from the asymmetry upper threshold detector in the asymmetry upper threshold circuit **410** of FIG. **4B**. The modification includes connecting the channel of an NMOS transistor **N2u** between the source terminal of the NMOS transistor **N1u** and the power supply voltages **VSSH**. The gate terminal of the NMOS transistor **N2u** is biased at an N-type “Middle Bias” voltage, whereby causing the NMOS transistor **N2u** to function as a resistive element. The asymmetry upper threshold detector in the asymmetry upper threshold circuit **810** of FIG. **8A** has an upper threshold voltage $V_{TH} + \Delta V_u$, which is larger than the upper threshold voltage V_{TH} of the asymmetry upper threshold detector in the asymmetry upper threshold circuit **410** of FIG. **4B** by a positive amount of ΔV_u .

[0095] FIG. **8B** is a schematic diagram of an asymmetry lower threshold circuit, in accordance with some embodiments. The asymmetry lower threshold detector in the asymmetry lower threshold circuit **820** of FIG. **8B** is modified from the asymmetry lower threshold detector in the asymmetry lower threshold circuit **420** of FIG. **4B**. The modification includes connecting the channel of a PMOS transistor **P2d** between the source terminal of the PMOS transistor **P1d** and the power supply voltages **VDDL**. The gate terminal of the PMOS transistor **P2d** is biased at a P-type “Middle Bias” voltage, whereby causing the PMOS transistor **P2d** to function as a resistive element. The asymmetry lower threshold detector in the asymmetry lower threshold circuit **820** of FIG. **8B** has a lower threshold voltage $V_{TL} - \Delta V_d$, which is smaller than the lower threshold voltage V_{TL} of the asymmetry lower threshold detector in the asymmetry lower threshold circuit **420** of FIG. **4B** by a positive amount of ΔV_d .

[0096] In some embodiments, the input buffer circuit **400B** of FIG. **4B** is modified to obtain an improved input buffer circuit. The modification includes replacing the asymmetry upper threshold

circuit **410** of FIG. **4B** with the asymmetry upper threshold circuit **810** of FIG. **8A** and replacing the asymmetry lower threshold circuit **420** of FIG. **4B** with the asymmetry lower threshold circuit **820** of FIG. **8B**. After the modification, the improved input buffer circuit has an enlarged hysteresis window $(V_{TH} + \Delta V_u) - (V_{TL} - \Delta V_d)$, which is a positive amount of $\Delta V_u + \Delta V_d$.

[0097] An aspect of the present disclosure relates to an integrated circuit. The integrated circuit includes an asymmetry upper threshold circuit having an upper-branch output node configured to output an upper-branch enabling signal generated from an input voltage signal, an asymmetry lower threshold circuit having a lower-branch output node configured to output a lower-branch enabling signal generated from the input voltage signal, and a control circuit configured to change an output voltage signal from a first voltage level to a second voltage level in response to a condition that a logic level of the upper-branch enabling signal and a logic level of the lower-branch enabling signal are changed consecutively. The asymmetry upper threshold circuit is configured to set the upper-branch enabling signal to a first upper-branch logic level in response to the input voltage signal rising across a main upper threshold, and configured to set the upper-branch enabling signal to a second upper-branch logic level in response to the input voltage signal falling across a supplementary upper threshold. The supplementary upper threshold is higher than the main upper threshold. The asymmetry lower threshold circuit is configured to set the lower-branch enabling signal to a first lower-branch logic level in response to the input voltage signal falling across a main lower threshold, and configured to set the lower-branch enabling signal to a second lower-branch logic level in response to the input voltage signal rising across a supplementary lower threshold. The supplementary lower threshold is lower than the main lower threshold.

[0098] Another aspect of the present disclosure relates to a method. The method includes generating an upper-branch enabling signal based on comparing an input voltage signal with either a main upper threshold or a supplementary upper threshold, generating a lower-branch enabling signal based on comparing the input voltage signal with either a main lower threshold or a supplementary lower threshold, and changing an output voltage signal from a first voltage level to a second voltage level in response to the logic level of the upper-branch enabling signal and the logic level of the lower-branch enabling signal are changed consecutively. Generating the upper-branch enabling signal includes changing a logic level of the upper-branch enabling signal to a first upper-branch logic level in response to the input voltage signal rising across the main upper threshold, and changing the logic level of the upper-branch enabling signal to a second upper-branch logic level in response to the input voltage signal falling across the supplementary upper threshold. The supplementary upper threshold is higher than the main upper threshold. Generating the lower-branch enabling signal includes changing a logic level of the lower-branch enabling signal to a first lower-branch logic level in response to the input voltage signal falling across the main lower threshold, and changing the logic level of the lower-branch enabling signal being set to a second lower-branch logic level in response to the input voltage signal rising across the supplementary lower threshold. The supplementary lower threshold is lower than the main lower threshold.

[0099] Another aspect of the present disclosure still relates to an integrated circuit. The integrated circuit includes an asymmetry upper threshold circuit configured to output an upper-branch enabling signal at an upper-branch output node, an asymmetry lower threshold circuit configured to output a lower-branch enabling signal at a lower-branch output node, a first switch configured to be controlled with the upper-branch enabling signal from the upper-branch output node, and a second switch configured to be controlled with the lower-branch enabling signal from the lower-branch output node. Each of the asymmetry upper threshold circuit and the asymmetry lower threshold circuit includes a first-type transistor having a gate terminal configured as an input terminal and having a drain terminal thereof configured as either the upper-branch output node or the lower-branch output node, and a plurality of second-type transistors having channels thereof serially connected between a voltage node and the drain terminal of the first-type transistor. The second-

type transistors have gate terminals thereof all connected to the gate terminal of the first-type transistor.

[0100] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated circuit comprising: an asymmetry upper threshold circuit having an upper-branch output node configured to output an upper-branch enabling signal generated from an input voltage signal, wherein the asymmetry upper threshold circuit is configured to set the upper-branch enabling signal to a first upper-branch logic level in response to the input voltage signal rising across a main upper threshold, and configured to set the upper-branch enabling signal to a second upper-branch logic level in response to the input voltage signal falling across a supplementary upper threshold, and wherein the supplementary upper threshold is higher than the main upper threshold; an asymmetry lower threshold circuit having a lower-branch output node configured to output a lower-branch enabling signal generated from the input voltage signal, wherein the asymmetry lower threshold circuit is configured to set the lower-branch enabling signal to a first lower-branch logic level in response to the input voltage signal falling across a main lower threshold, and configured to set the lower-branch enabling signal to a second lower-branch logic level in response to the input voltage signal rising across a supplementary lower threshold, and wherein the supplementary lower threshold is lower than the main lower threshold; and a control circuit configured to change an output voltage signal from a first voltage level to a second voltage level in response to a condition that a logic level of the upper-branch enabling signal and a logic level of the lower-branch enabling signal are changed consecutively.
2. The integrated circuit of claim 1, wherein: the control circuit is configured to change the output voltage signal from the first voltage level to the second voltage level in response to a condition that the logic level of the upper-branch enabling signal is changed after the logic level of the lower-branch enabling signal is changed.
3. The integrated circuit of claim 1, wherein: the control circuit is configured to change the output voltage signal from the second voltage level to the first voltage level in response to a condition that the logic level of the upper-branch enabling signal is changed before the logic level of the lower-branch enabling signal is changed.
4. The integrated circuit of claim 1, wherein the asymmetry upper threshold circuit comprises: an NMOS transistor having a drain terminal thereof connected to the upper-branch output node; a plurality of PMOS transistors having channels thereof serially connected between a first voltage node and the drain terminal of the NMOS transistor; and where gate terminals of the NMOS transistor and the PMOS transistors are connected together.
5. The integrated circuit of claim 4, wherein the asymmetry upper threshold circuit further comprises: a transmission gate connected between the drain terminal of the NMOS transistor and the drain terminal of one of the PMOS transistors, the transmission gate having a conducting state thereof controlled with a threshold-selection signal.
6. The integrated circuit of claim 1, wherein the asymmetry lower threshold circuit comprises: a PMOS transistor having a drain terminal thereof connected to the upper-branch output node; a plurality of NMOS transistors having channels thereof serially connected between a second voltage

node and the drain terminal of the PMOS transistor; and where gate terminals of the PMOS transistor and the NMOS transistors are connected together.

7. The integrated circuit of claim 6, wherein the asymmetry upper threshold circuit further comprises: a transmission gate connected between the drain terminal of the PMOS transistor and the drain terminal of one of the NMOS transistors, the transmission gate having a conducting state thereof controlled with a threshold-selection signal.

8. The integrated circuit of claim 1, further comprising: a threshold-selection signal circuit having a delay circuit therein and configured to generate a threshold-selection signal in response to a change of the output voltage signal.

9. The integrated circuit of claim 1, wherein the control circuit comprises: a first switch configured to be controlled with the upper-branch enabling signal from the upper-branch output node; a second switch configured to be controlled with the lower-branch enabling signal from the lower-branch output node; and a buffer output node at an electric connection between the first switch and the second switch.

10. The integrated circuit of claim 9, wherein: the first switch is electrically connected between a first voltage node and the buffer output node; and the second switch is electrically connected between the buffer output node and a second voltage node.

11. The integrated circuit of claim 9, further comprising: a regenerative circuit electrically coupled to the buffer output node and configured to maintain a voltage at the buffer output node during a time period when the first switch is at a disconnected state and the second switch is at a disconnected state.

12. A method comprising: generating an upper-branch enabling signal based on comparing an input voltage signal with either a main upper threshold or a supplementary upper threshold, wherein generating the upper-branch enabling signal includes changing a logic level of the upper-branch enabling signal to a first upper-branch logic level in response to the input voltage signal rising across the main upper threshold, and changing the logic level of the upper-branch enabling signal to a second upper-branch logic level in response to the input voltage signal falling across the supplementary upper threshold, and wherein the supplementary upper threshold is higher than the main upper threshold; generating an lower-branch enabling signal based on comparing the input voltage signal with either a main lower threshold or a supplementary lower threshold, wherein generating the lower-branch enabling signal includes changing a logic level of the lower-branch enabling signal to a first lower-branch logic level in response to the input voltage signal falling across the main lower threshold, and changing the logic level of the lower-branch enabling signal being set to a second lower-branch logic level in response to the input voltage signal rising across the supplementary lower threshold, and wherein the supplementary lower threshold is lower than the main lower threshold; and changing an output voltage signal from a first voltage level to a second voltage level in response to the logic level of the upper-branch enabling signal and the logic level of the lower-branch enabling signal are changed consecutively.

13. The method of claim 12, wherein changing the output voltage signal comprises: changing the output voltage signal from the first voltage level to the second voltage level in response to a condition that the logic level of the upper-branch enabling signal is changed after the logic level of the lower-branch enabling signal is changed.

14. The method of claim 12, wherein changing the output voltage signal comprises: changing the output voltage signal from the second voltage level to the first voltage level in response to a condition that the logic level of the upper-branch enabling signal is changed before the logic level of the lower-branch enabling signal is changed.

15. The method of claim 12, further comprises: controlling a first switch with the upper-branch enabling signal, wherein the first switch is electrically connected between a first voltage node and a buffer output node, wherein the first switch is configured to be controlled by the upper-branch enabling signal; and controlling a second switch with the lower-branch enabling signal, wherein the

- second switch is electrically connected between the buffer output node and a second voltage node.
- 16.** An integrated circuit comprising: an asymmetry upper threshold circuit configured to output an upper-branch enabling signal at an upper-branch output node; an asymmetry lower threshold circuit configured to output a lower-branch enabling signal at a lower-branch output node; a first switch configured to be controlled with the upper-branch enabling signal from the upper-branch output node; and a second switch configured to be controlled with the lower-branch enabling signal from the lower-branch output node, wherein each of the asymmetry upper threshold circuit and the asymmetry lower threshold circuit comprises, a first-type transistor having a gate terminal configured as an input terminal and having a drain terminal thereof configured as either the upper-branch output node or the lower-branch output node, and a plurality of second-type transistors having channels thereof serially connected between a voltage node and the drain terminal of the first-type transistor, the second-type transistors having gate terminals thereof all connected to the gate terminal of the first-type transistor.
- 17.** The integrated circuit of claim 16, wherein: the first-type transistor in the asymmetry upper threshold circuit is an NMOS transistor; and the second-type transistors in the asymmetry upper threshold circuit are PMOS transistor.
- 18.** The integrated circuit of claim 16, wherein: the first-type transistor in the asymmetry lower threshold circuit is a PMOS transistor; and the second-type transistors in the asymmetry lower threshold circuit are NMOS transistors.
- 19.** The integrated circuit of claim 16, further comprising: a linear switch connected between the drain terminal of the first-type transistor and the drain terminal of one of the second-type transistors, the linear switch having a conducting state thereof controlled with a threshold-selection signal.
- 20.** The integrated circuit of claim 19, further comprising: a threshold-selection signal circuit having a delay circuit therein and configured to generate the threshold-selection signal in response to an output voltage signal from a first voltage level to a second voltage level.
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